Adaptive Tutoring Beyond Classical RAG: Conditional Retrieval in a Repository-Grounded Multi-Agent Study

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***Abstract*— Retrieval-augmented generation grounds tutor responses in external evidence, but effective tutoring also requires learner diagnosis, scaffold sequencing, and rubric alignment. We study adaptive tutoring as an *architecture-selection* problem: should retrieval be the default controller, or should it be invoked only when the task regime demands explicit evidence? Under shared freeze-time controls (identical corpus, chunking, reranker, prompt skeleton, and budgets), we compare four repository-grounded configurations—retrieval-first classical RAG, non-RAG multi-agent coordination, a single-agent baseline, and our proposed conditional-retrieval hybrid—all driven by the same small open-weight model (Qwen3.5-4B, thinking budget 512). On fully converged TutorEval [1] ($n = 834$, identical examples across architectures), the hybrid achieves point-estimate improvements of $+3.3\%$ Token-F1 ($0.1115 \rightarrow 0.1152$), $+5.0\%$ grounded overlap ($0.2812 \rightarrow 0.2953$), -5.4% latency, and -12.2% tokens relative to retrieval-first, while matching or exceeding rubric coverage and key-point recall. On partial EduBench [2], architectures processed non-overlapping subsets (4.0%–13.3% coverage), which precludes valid cross-architecture comparison; we therefore report EduBench only as within-subset trajectory evidence. The controlled TutorEval results show that *conditional* retrieval, paired with lightweight specialist preparation, can match retrieval-first quality at strictly lower cost and motivate a view of adaptive tutoring as a regime-dependent architecture-selection problem rather than a uniformly retrieval-centered one.**

I. Introduction

A student asks a tutoring system, "What does this concept mean?" The right answer is a short explanation grounded in source material—essentially a retrieval problem. Another student asks, "Why is my answer wrong?" That task is different: the tutor must decide whether to explain directly, offer a scaffolded hint, align feedback to a rubric, or deliberately withhold part of the answer to preserve the learning opportunity. What looks like a retrieval problem has become an instructional decision problem.

This paper studies adaptive tutoring through that lens. We frame tutoring as a *regime-dependent architecture-selection problem* and ask a focused systems question: should retrieval be the default controller for every tutoring interaction, or should it be invoked only when the task regime demands explicit evidence? To answer, we build a repository-grounded multi-agent framework and compare four configurations—retrieval-first classical RAG, non-RAG multi-agent coordination, a single-agent baseline, and a proposed hybrid with *conditional* retrieval—under shared freeze-time controls.

Retrieval-augmented generation (RAG) grounds model outputs in external evidence [3] and remains a strong default for fact-seeking question answering. Tutoring, however, requires more than grounded recall. A useful tutor must infer learner state, choose how much help to provide, follow pedagogical constraints, and decide when instructional strategy should take priority over direct answer delivery. The question is therefore not whether retrieval helps, but *when* retrieval is sufficient and when additional tutoring control matters.

Recent education-centered benchmarks make that distinction harder to ignore. EduBench and TutorEval evaluate not only correctness but pedagogy, adaptivity, reasoning, and grounded tutoring behavior [1], [2], [4], and a growing evaluation literature emphasizes open-ended pedagogical quality, adaptivity, and safety over answer accuracy alone [5]–[10]. In these settings, a system can be factually correct yet still tutor poorly by giving the wrong level of help, missing rubric criteria, or failing to adapt.

We use *orchestration* to denote the additional control required for such cases: diagnosis, planning, routing, rubric use, critique, and intervention *before or during* answer generation. Our framework makes this control explicit through regime-aware routing and conditional retrieval. From a systems-and-cybernetics perspective, the router acts as a lightweight supervisory controller over retrieval and pedagogical sub-policies, deciding when evidence grounding should dominate and when instructional coordination should dominate. Rather than assuming one architecture governs every interaction, we compare the four configurations above under identical infrastructure. Retrieval is available when evidence is needed, but it is not treated as the controller by default.

Contributions. (i) We formalize adaptive tutoring as architecture selection conditioned on a lightweight regime router and operationalize it in a repository-grounded framework that instantiates classical RAG, non-RAG multi-agent, single-agent, and hybrid conditional-retrieval pipelines on a shared substrate, with the router serving as a supervisory controller over retrieval and pedagogical sub-policies. (ii) We report a controlled freeze-time comparison in which TutorEval is fully converged on identical examples ($n = 834$) and hybrid achieves the best point-estimate quality-efficiency tradeoff. (iii) We provide explicit, reviewer-safe semantics for subset-based EduBench trajectory evidence, profile-gated metrics, and retrieval-sensitive grounding scores—avoiding the cross-subset comparisons that would inflate claims. Code, prompts, and the frozen snapshot schema will be released upon publication.

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II. Related Work

Our work sits at the intersection of three lines: educational tutoring benchmarks, retrieval control for RAG, and multi-agent orchestration. We highlight where each line stops and why the boundary between them motivates the experiments in this paper.

Educational tutoring benchmarks. Recent educational benchmarks move the evaluation target beyond answer correctness toward pedagogy, adaptivity, and instructional safety. EduBench spans nine tutoring scenarios and scores educational utility dimensions such as adaptation and pedagogical application [2]. TutorEval operationalizes chapter-grounded science tutoring with keypoint-based evaluation [1], [4]. MathTutorBench, TutorBench, and related evaluation frameworks sharpen open-ended pedagogical assessment and tutor-specific safety [5]–[10]. These benchmarks crystallize the gap our paper targets: a tutor can be factually correct yet still tutor poorly by giving the wrong level of help, neglecting rubric criteria, or responding unsafely, and optimizing for answer correctness alone will not recover that behavior.

RAG and agentic retrieval control. Classical RAG conditions generation on retrieved passages [3] and is a strong default for fact-seeking QA. Subsequent work shows that retrieval quality depends on the *control policy* around it: Self-RAG trains reflection tokens that decide when to retrieve and when to critique an answer [11]; CRAG adds a lightweight retrieval evaluator that triggers corrective actions when retrieved evidence is poor [12]; and adaptive RAG variants have been studied for short-form open-domain QA [13]. A parallel evaluation literature diagnoses retrieval-heavy systems through answer quality, citation behavior, and benchmarking infrastructure [14]–[17]. In tutoring, however, many queries are dominated by diagnosis, hinting, or rubric-following rather than evidence-seeking, and long educational contexts can make naive "retrieve by default" behavior brittle [18], [19]. Unlike Self-RAG's learned reflection tokens or CRAG's retrieval evaluator, our router is a training-free, *sub-model-call* heuristic over lexical cues and dataset priors (§III), which keeps latency and token costs predictable in resource-constrained tutoring deployments—a positioning orthogonal to, rather than competing with, learned adaptive RAG.

Multi-agent tutoring and coordination-heavy control. A broad multi-agent literature argues that explicit role specialization helps with planning, critique, and decomposition in complex workflows [20]–[23]. In retrieval settings, collaborative agents have also been proposed to improve search and reasoning over evidence [24], [25]. Closer to our problem, recent tutoring systems including AgentTutor, IntelliCode, and CogEvo-Edu instantiate specialized educational roles such as learner modeling, multi-turn teaching, and collaborative tutoring [26]–[28]. These designs motivate tutoring-specific orchestration, but they do not isolate the architectural question we study: *when* should evidence retrieval be the primary controller, and when should it remain a subordinate capability inside a broader tutoring policy?

Positioning. Prior work supplies the ingredients—educational benchmarks that reward pedagogy, retrieval methods that improve grounding, and multi-agent frameworks that improve coordination—but leaves the boundary between them under-specified. Our contribution is a repository-grounded comparison that makes this boundary explicit by testing retrieval-first, coordination-heavy, and hybrid conditional-retrieval control under shared infrastructure and freeze-time conditions.

III. Methodology

We now describe the framework and the experimental protocol used for the frozen comparison.

The framework implements five execution families—classical RAG, agentic RAG, non-RAG multi-agent, single-agent baseline, and our proposed conditional-retrieval hybrid (`hybrid_fast`). A lightweight router selects the family and sets execution flags for each example (Fig. 1). The frozen comparison in this paper uses four families with comparable outputs; the implemented `agentic_rag` path is excluded because parity-controlled outputs were unavailable at freeze time (see §VI). The design principle is *controlled adaptivity*: invoke retrieval only when an instance needs external evidence, and otherwise spend the saved budget on pedagogical coordination.

A. Regime-aware routing

For each example x, the router computes five heuristic scores $s(x) \in [0,1]^5$ from normalized keyword cues (evidence, coordination, rubric, planning, tutoring), augmented with dataset priors (e.g., TutorEval→*adaptive tutoring*, HotpotQA→*evidence grounded*). An optional logistic-regression classifier can override the predicted family when a trained router is loaded; the frozen runs use the heuristic+prior fallback throughout. The selected family and the retrieval gate for `hybrid_fast` follow:

$$\text{family}(x) = \begin{cases} \text{CLS-RAG} & s_e(x) \geq \tau_e^{\text{cls}},\ s_c(x) < 0.35 \\ \text{AGN-RAG} & s_e(x) \geq 0.35,\ s_c(x) \geq \tau_c \\ \text{NON-RAG} & s_c(x) \geq 0.55,\ s_e(x) < 0.28 \\ \text{HYB-FAST} & \text{otherwise}, \end{cases} \tag{1}$$

where s_e, s_c are the evidence and coordination scores and $(\tau_e^{\text{cls}}, \tau_c) = (0.52, 0.50)$. Within `hybrid_fast`, the retrieval gate is

$$\text{require_retrieval}(x) = \mathbb{1}[s_e(x) \geq 0.35 \ \vee \ \text{regime}(x) = \text{EG}], \tag{2}$$

where EG denotes the *evidence-grounded* regime. A fallback retrieval call is issued when the primary path returned *no* chunks but $s_e(x) \geq 0.45$ (a small safety margin above the gate), and the tutor is re-invoked with the recovered evidence. These thresholds are the production values shipped with the repository; we report sensitivity as a planned ablation in §IV.

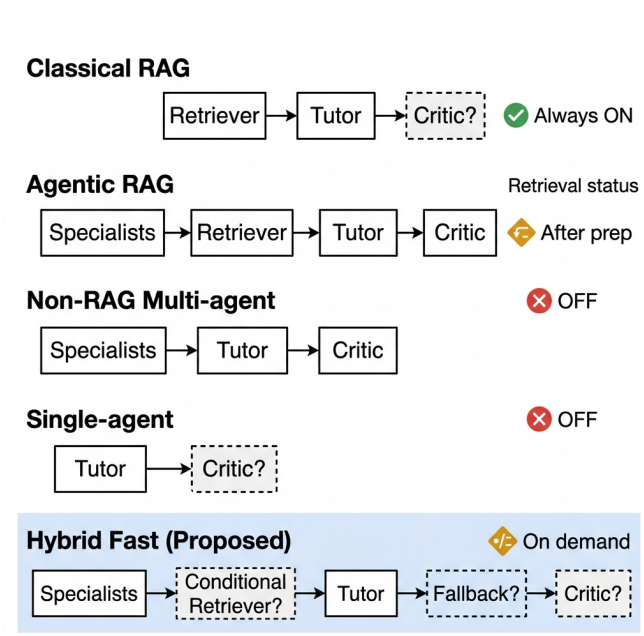

Fig. 1. Repository-grounded execution families. Solid boxes are always-active modules; dashed boxes are optional. The proposed `hybrid_fast` path (highlighted) combines parallel specialist coordination with conditional retrieval and a single fallback.

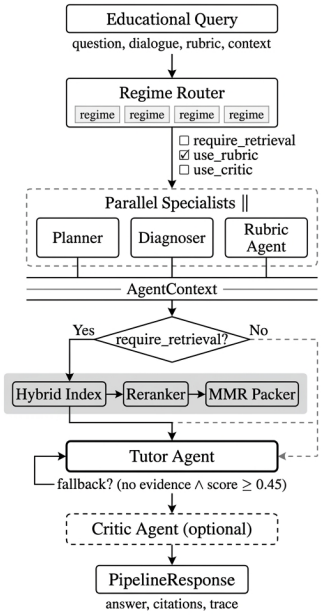

Fig. 2. Overview of the `hybrid_fast` pipeline. Solid arrows are always-active paths; dashed arrows are conditionally activated by the router (Eq. 2) and by issue-triggered critic revision. Three parallel specialists prepare the shared context; the retrieval stack activates only when `require_retrieval` is set; a single fallback re-queries when evidence need is high ($s_e \geq 0.45$) but no chunks were returned.

B. Specialist agents and shared context

All execution families operate on a shared `AgentContext` carrying the example, route decision, student state, search queries, rubric summary, retrieved chunks, and draft answer. Specialists are lightweight and launched in parallel where possible: the **planner** emits up to three search queries tailored to the regime; the **diagnoser** infers learner level, goals, style, and likely misconceptions; the **rubric agent** compiles a checklist from rubric items; the **tutor** drafts the final response using retrieved evidence when available; and an optional **critic** inspects the draft for concrete issues (missing evidence markers, unmet rubric items, length mismatches) and revises only when such issues are detected. All agents share the same small model backbone and shared budget policy, so differences across families reflect *structural* rather than capacity differences.

C. Hybrid retrieval, reranking, and context packing

The retrieval stack is intentionally lightweight. A hybrid index combines word-level TF–IDF ($\lambda_w = 0.58$), character n-gram TF–IDF ($\lambda_c = 0.17$), and an optional truncated-SVD latent projection ($\lambda_\ell = 0.25$); the weighted similarity is boosted by query–title token overlap. A six-feature logistic reranker (base score, token/title overlap, phrase match, numeric overlap, brevity) reorders the candidates, and a maximum marginal relevance [29] packer with $\lambda_{\text{MMR}} = 0.68$ selects up to four chunks within a 6,000-character context budget.

D. End-to-end hybrid procedure

Fig. 2 summarizes the `hybrid_fast` pipeline. The router classifies each example via Eq. (1) and sets execution flags; parallel specialists (planner, diagnoser, rubric agent) prepare the shared context; retrieval activates only when Eq. (2) is satisfied; the tutor drafts an answer; a single fallback retrieval call is triggered when no chunks were obtained and $s_e(x) \geq 0.45$; the critic revises iff concrete issues are detected.

E. Evaluation protocol

Metric handling is *profile-aware* and enforced in code, so cross-profile metrics report structural zeros rather than post-hoc estimates. Shared metrics include exact match, token-level F1, choice accuracy (when applicable), citation coverage, grounded overlap, rubric coverage, adaptivity, retrieval-document recall, latency, token counts, and agent count. EduBench additionally reports JSON-format compliance and score alignment against consensus-style reference scores extracted by the dataset adapter, plus a 12-dimension scenario/factual/pedagogical rubric. TutorEval additionally reports keypoint precision/recall and chapter grounding. Two caveats matter for interpretation: (i) *grounded overlap* is structurally zero for non-retrieval architectures and should not be used to rank retrieval- vs. non-retrieval systems; (ii) *citation coverage* returns a perfect fallback (1.0) when no chunks are retrieved, so it should be read alongside the retrieval count, not in isolation.

IV. Experimental Results

This section presents the experimental design and the frozen-snapshot results. All quantitative claims are limited to the frozen comparison (converged TutorEval endpoint, architecture-specific partial EduBench subsets, four configu-

rations with comparable outputs) and should not be read as benchmark-wide rankings or classroom-effectiveness claims.

A. Setup

Compared methods. The frozen comparison uses four configurations (Fig. 1): (a) classical RAG (retrieval followed by tutor); (b) non-RAG multi-agent (planner + diagnoser + rubric specialists, no retrieval); (c) single-agent (tutor only); and (d) the proposed `hybrid_fast` (Fig. 2) with parallel specialist preparation, conditional retrieval, and fallback. The implemented `agentic_rag` path is excluded only because parity-controlled frozen outputs were unavailable at freeze time; the design remains part of the framework and is reserved for ablation work rather than discarded.

Model and hyperparameters. All architectures use the same open-weight backbone (QWEN3.5-4B) with a fixed reasoning/"thinking" budget of 512 tokens via an OpenAI-compatible endpoint, temperature 0.15, and a per-response cap of 900 tokens. Retrieval settings: word/character/latent weights $(0.58, 0.17, 0.25)$, latent dim 128, top-$k = 8$, final-$k = 4$, chunk size 220 with 40-token overlap, MMR $\lambda = 0.68$. Budget caps: at most 3 retrieval queries, 4 tool calls, 4 agents, one critic round per example, and a $6{,}000$-character context window. The configuration also records a 12-second latency budget target, but in the frozen code path this value is a planning parameter rather than an enforced runtime cutoff; the reported latencies are post hoc end-to-end measurements from the executed pipelines. The router thresholds are those in §III.

Benchmarks. EduBench spans nine educational scenarios and twelve teacher- and student-facing aspects [2]. TutorEval assesses chapter-grounded science tutoring with keypoint-oriented evaluation [1]. Transfer benchmarks [18], [20], [30]–[34] are supported by the framework but retained here only for narrower mechanism claims.

Parity controls. All systems share the same indexed corpus, chunking policy, retrieval budget, reranker, and packer (*retrieval parity*); the same prompt skeleton, student metadata, and rubric anchors (*prompt parity*); and the same small-model backbone (*capacity parity*). Token counts, tool calls, retrieved chunks, agent count, and latency are logged per run (*budget transparency*). TutorEval is fully converged ($834/834$ across all four architectures, identical examples). EduBench at freeze time is a partial trajectory with architecture-specific coverage ranging from 4.0% to 13.3% (Table I); the EduBench subsets are *not* intersected across architectures, which is the single most important constraint on interpretation below.

B. Quality-efficiency comparison

Table II summarizes the frozen snapshot. On the fully converged TutorEval endpoint ($n = 834$, identical examples across all architectures), `hybrid_fast` attains the best point estimates on Token-F1, grounded overlap, rubric coverage, and keypoint recall, while simultaneously reducing mean latency and token usage. Relative to classical RAG on TutorEval, hybrid delivers $+3.3\%$ Token-F1 ($0.1115 \rightarrow 0.1152$), $+5.0\%$ grounded overlap ($0.2812 \rightarrow 0.2953$), -5.4% latency ($51{,}903 \rightarrow 49{,}113$ ms), and -12.2% total tokens ($3{,}975 \rightarrow 3{,}490$). Non-RAG and single-agent families reach respectable rubric coverage but trail on Token-F1 and, by design, on grounded overlap. We report these as point estimates from a single converged run; the small absolute magnitudes (~ 0.004 Token-F1) mean that without paired bootstrap confidence intervals on per-example scores, none of these deltas should be treated as formally significant (see §VI).

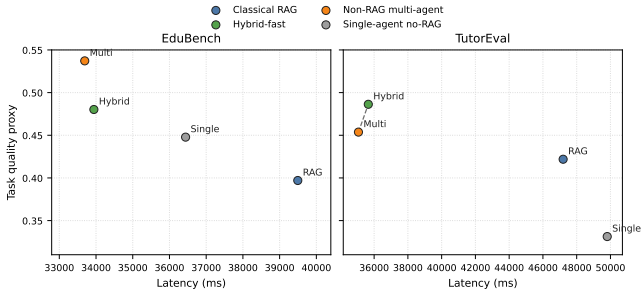

Fig. 3. Frozen quality–latency geometry. TutorEval points are converged endpoints over identical examples; EduBench points are partial-trajectory snapshots over non-overlapping subsets and are reported only as within-subset descriptions.

For EduBench, we report subset-based trajectory evidence only. Coverage ranges from 4.0% to 13.3%, and the four architectures were evaluated on *different*, non-overlapping processed-success subsets; cross-architecture EduBench comparisons would therefore confound selection effects with architectural effects. Within-subset patterns show coordination-heavy routes at 0.954–0.955 rubric coverage with lower latency and tokens, classical RAG with non-zero grounded overlap (0.338), and low EduAlign across all architectures (classical RAG highest at 0.242). Exact match, Token-F1, and choice accuracy are structurally 0.0 in the EduBench profile because public rows typically ship no direct gold answers. These EduBench values support within-subset description but not cross-architecture ranking.

The controlled TutorEval results support a qualitative claim that we treat as the paper's main take-away: *conditional* retrieval can match retrieval-first quality at strictly lower cost when architectures are compared on identical examples. Retrieval remains necessary for evidence-sensitive tutoring behavior, but unconditional retrieval-first control is not uniformly optimal in quality-efficiency space.

C. Tradeoff geometry and ablation outlook

Fig. 3 shows the frozen quality–latency geometry. On TutorEval (converged), `hybrid_fast` sits on the upper-left of the front (higher quality, lower latency) relative to classical RAG, non-RAG, and single-agent. On EduBench (partial), the ordering is consistent with the parity-controlled caveat above but should not be read as an endpoint ranking.

The frozen snapshot evaluates complete architecture configurations rather than isolated ablation variants. Four repository-aligned ablations are planned for the camera-ready revision: (1) incorporating the `agentic_rag` path to assess

TABLE I

RUN COMPLETION AT FREEZE TIME. EDUBENCH SESSIONS ARE STILL ACTIVE AND SHOULD BE READ AS PARTIAL TRAJECTORY SNAPSHOTS; TUTOREVAL SESSIONS ARE CONVERGED. METRICS IN SUBSEQUENT TABLES ARE COMPUTED ONLY OVER EACH ARCHITECTURE'S SUCCESSFULLY COMPLETED SUBSET.

Dataset	Architecture	Success	Processed	Total	Coverage (%)	Pending	Failed
EduBench	classical_rag	682	682	16956	4.02	16274	0
EduBench	hybrid_fast	2254	2257	16956	13.31	14699	3
EduBench	non_rag_multi_agent	2104	2106	16956	12.42	14850	2
EduBench	single_agent_no_rag	1338	1344	16956	7.93	15612	6
TutorEval	classical_rag	834	834	834	100.00	0	0
TutorEval	hybrid_fast	834	834	834	100.00	0	0
TutorEval	non_rag_multi_agent	834	834	834	100.00	0	0
TutorEval	single_agent_no_rag	834	834	834	100.00	0	0

TABLE II

FROZEN QUALITY-EFFICIENCY SNAPSHOT. EDUBENCH VALUES ARE SUBSET-BASED PARTIAL-TRAJECTORY MEANS OVER SUCCESSFULLY COMPLETED EXAMPLES ONLY; TUTOREVAL VALUES ARE CONVERGED ENDPOINTS. HIGHER IS BETTER EXCEPT LATENCY AND TOKENS.

Dataset	Architecture	Cov. (%)	Tok-F1	Ground.	Rub.	EduAlign	KeyRec	Lat. (ms)	Tokens
EduBench	classical_rag	4.02	0.0000	0.3384	0.7987	0.2419	–	48170.2	3871.4
EduBench	hybrid_fast	13.31	0.0000	0.0000	0.9553	0.0302	–	29950.7	1757.8
EduBench	non_rag_multi_agent	12.42	0.0000	0.0000	0.9541	0.0347	–	32327.2	1790.7
EduBench	single_agent_no_rag	7.93	0.0000	0.0000	0.7345	0.0933	–	50282.8	3041.5
TutorEval	classical_rag	100.00	0.1115	0.2812	0.9773	–	0.8901	51903.1	3974.7
TutorEval	hybrid_fast	100.00	**0.1152**	**0.2953**	**0.9902**	–	**0.8974**	**49113.4**	**3490.3**
TutorEval	non_rag_multi_agent	100.00	0.0898	0.0000	0.9810	–	0.8672	51048.0	5959.3
TutorEval	single_agent_no_rag	100.00	0.0865	0.0000	0.9688	–	0.8423	55257.3	6466.2

whether specialist preparation *before* retrieval helps retrieval-intensive tutoring; (2) disabling the conditional retrieval gate in the hybrid pathway to separate route-specific retrieval effects from planner/critic effects; (3) enabling retrieval for the strongest non-RAG multi-agent baseline to distinguish grounding failures from coordination issues; and (4) removing the critic phase to isolate the value of post-hoc revision. Threshold sensitivity ($\tau_e \in [0.30, 0.50]$) and router-classifier vs. heuristic-only comparisons are enabled by the repository but not reported here. The present paper commits to the stronger, narrower claim supported by the frozen TutorEval endpoint; the planned ablations are designed to stress-test the mechanism, not to rescue it.

V. DISCUSSION

Three observations structure the take-away. *First*, retrieval is necessary for evidence traceability: only retrieval-enabled architectures achieve non-zero grounded overlap in the TutorEval profile, and the gap between classical RAG and non-RAG on Token-F1 (0.1115 vs. 0.0898) is consistent with the hypothesis that evidence grounding carries real weight on science-tutoring queries. *Second*, retrieval is not sufficient: `hybrid_fast` adds parallel specialist preparation on top of conditional retrieval and improves on classical RAG across Token-F1, grounded overlap, rubric coverage, and keypoint recall, at lower latency and token cost. This is the paper's sharpest finding and the one that most directly contradicts a pure retrieval-first view. *Third*, coordination alone is brittle: non-RAG multi-agent matches hybrid on rubric coverage but underperforms on Token-F1 and grounded overlap, suggesting that orchestration without access to evidence cannot substitute for selective retrieval when the task does require it.

These patterns argue for treating adaptive tutoring as an architecture-selection problem rather than a retrieval-parameter problem. Viewed through an SMC lens, the central design object is a supervisory policy that coordinates these sub-processes under uncertainty and resource limits. Conditional retrieval is one answer; richer controllers with calibrated uncertainty, explicit per-example budgets, and human override hooks are a natural next step.

VI. LIMITATIONS

We emphasize four constraints. **(i) Partial EduBench.** EduBench remains a partial snapshot (4.0%–13.3% coverage) with architecture-specific subsets; we therefore report EduBench only as within-subset trajectory evidence and draw no cross-architecture ordering claims from it. **(ii) No statistical testing.** We report point estimates from a single converged TutorEval run; paired bootstrap confidence intervals and permutation tests on per-example scores are required to promote the observed TutorEval deltas to formally significant claims, and we flag this explicitly rather than implying significance. **(iii) Profile-gated metrics.** Exact match, Token-F1, and choice accuracy are structurally 0.0 in the EduBench consensus profile; grounded overlap is structurally 0.0 when retrieval is disabled; citation coverage returns a perfect fallback when no chunks are retrieved. These values should be read alongside, not in place of, the profile-appropriate metrics. **(iv) Scope.** The study uses a single small backbone (Qwen3.5-4B, thinking budget 512), a single corpus, and English-language benchmarks, and it evaluates benchmark outputs rather than downstream learning gains. Generalization to larger models, multilingual data,

and classroom deployment is explicitly not tested here.

VII. Conclusion

We compared retrieval-first, coordination-heavy, and hybrid conditional-retrieval designs under shared freeze-time controls on a single shared small-model backbone. On fully converged TutorEval ($n = 834$, identical examples), the hybrid achieves the best point-estimate quality at lower latency and tokens ($+3.3\%$ Token-F1, $+5.0\%$ grounded overlap, -5.4% latency, -12.2% tokens vs. classical RAG), while coordination-heavy and single-agent baselines trail on the retrieval-sensitive metrics. Read together with the explicit EduBench caveats, the evidence supports framing adaptive tutoring as a *regime-dependent architecture selection problem*: retrieval remains essential for evidence-sensitive tasks but functions most effectively as a *conditional* capability inside a broader tutoring policy. The framework, prompts, and frozen snapshot schema will be released upon publication. Future work will fully converge EduBench on intersected subsets, add paired bootstrap confidence intervals, report the planned ablations, and extend to human evaluation and larger-backbone transfer.

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